Statistics 3515

Lecture Notes no *6*

Point Estimates for Variance Components in One Way Random ANOVA

We use the methods of moments approach to derive point estimates for the variance components:

1. To estimate σ_ε^2 we use the fact that

$$E(MS_E) = \sigma_\varepsilon^2$$

and therefore the method of moments estimator for σ_ε^2 is:

$$\widehat{\sigma_\varepsilon^2} = MS_E.$$

2. To estimate σ_τ^2 we use the fact that

$$E(MS_{Treatment}) = \sigma_\varepsilon^2 + n_0\sigma_\tau^2,$$

where $n_0 = n$ in a balanced design and

$$n_0 = \frac{1}{a-1}\left[\sum_{i=1}^{a} n_i - \frac{\sum_{i=1}^{a} n_i^2}{\sum_{i=1}^{a} n_i}\right],$$

in an unbalanced design.

Therefore, the method of moments estimator of the variance component due to treatments is:

$$\widehat{\sigma_\tau^2} = \left(MS_{Treatment} - MS_E\right)/n_0.$$

For our example we get from the ANOVA table in the SAS output:

$$\widehat{\sigma_\varepsilon^2} = 75.6$$

and

$$\widehat{\sigma_\tau^2} = 73.6.$$

Therefore,

$$\widehat{Var(Y_{ij})} = \widehat{\sigma^2_{Total}} = \widehat{\sigma_\varepsilon^2} + \widehat{\sigma_\tau^2} = 75.6 + 73.6 = 149.2$$

and the estimate for the proportion of variability due to the personnel officers (treatment) is

$$\widehat{\rho} = \frac{\widehat{\sigma_\tau^2}}{\widehat{\sigma_\varepsilon^2} + \widehat{\sigma_\tau^2}} = \frac{73.6}{149.2} = .49$$

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Confidence Intervals for Variance Components in One Way ANOVA

A $(1-\alpha)100\%$ confidence interval for σ_ε^2 is given by

$$\left(\frac{(N-a)MS_E}{\chi^2_{\alpha/2;N-a}}, \frac{(N-a)MS_E}{\chi^2_{1-\alpha/2;N-a}}\right).$$

From the Chi square tables in the book we get that a 95% confidence interval for σ_ε^2 is given by

$$(41.25; 180.86).$$

A $(1-\alpha)100\%$ confidence interval for

$$\rho = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma_e^2}$$

is given by

$$\left(\frac{L}{L+1}; \frac{U}{U+1}\right),$$

where

$$L = \frac{1}{n}\left(\frac{MS_{Treatment}}{MS_E} \cdot \frac{1}{F_{\alpha/2;a-1,N-a}} - 1\right)$$

and

$$U = \frac{1}{n}\left(\frac{MS_{Treatment}}{MS_E} \cdot \frac{1}{F_{1-\alpha/2;a-1,N-a}} - 1\right).$$

For our example:

$\alpha = .05$, $a-1 = 4$ and $N-a = 15$.

$F = \frac{MS_{Treatment}}{MS_{Error}} = 4.89$, $F_{.025;4,15} = 3.80$, $F_{.975;4,15} = 1/F_{.025;15,4} = 8.66$.

Therefore, $L = [(4.89/3.80) - 1]/4 = .0717105$ and

$U = [(4.89)(8.66) - 1]/4 = 10.33685$.

This implies that a 95% confidence interval for ρ is given by:

(.0669; .9112).

There are no exact confidence intervals for σ_τ^2. Several methods have been used to derive an approximate confidence interval for σ_τ^2. One confidence interval is presented in Section 13.6 that you might want to read at a later time. Here is a confidence interval based on a method described in Chapter 7, Section 2 of the book *The Analysis of Variance* by H. Scheffe, Wiley, 1959.

A $(1-\alpha)100\%$ confidence interval for σ_τ^2 is given by $(L; U)$, where

$$L = \frac{MS_E}{n}\left\{\frac{F}{F_{\alpha/2;a-1,\infty}} - 1 - \frac{F_{\alpha/2;a-1,N-a}}{F}\left(\frac{F_{\alpha/2;a-1,N-a}}{F_{\alpha/2;a-1,\infty}} - 1\right)\right\},$$

$$U = \frac{MS_E}{n}\left\{F_{\alpha/2;\infty,a-1}F - 1 + \frac{1}{F_{\alpha/2;N-a,a-1}F}\left(1 - \frac{F_{\alpha/2;\infty,a-1}}{F_{\alpha/2;N-a,a-1}}\right)\right\},$$

$$F = \frac{MS_{Treatment}}{MS_E},$$

and the appropriate tails of the F distribution are obtained from the tables in the book.

For a 95% confidence interval for σ_τ^2 in our example we have:

$F = 4.89$, $MS_E = 75.6$, $n = 4$, $F_{\alpha/2;a-1,N-a} = 3.80$, $F_{\alpha/2;a-1,\infty} = 2.79$, and therefore

$$L = \frac{75.6}{4}\left\{\frac{4.89}{2.79} - 1 - \frac{3.80}{4.89}\left(\frac{3.80}{2.79} - 1\right)\right\} = 8.91.$$

Now, $F_{\alpha/2;\infty,a-1} = 8.26$ and $F_{\alpha/2;N-a,a-1} = 8.66$. Therefore,

$$U = \frac{75.6}{4}\left\{(8.26)(4.89) - 1 + \frac{1}{(8.66)(4.89)}\left(1 - \frac{8.26}{8.66}\right)\right\} = 744.48.$$